CORNELL UNIVERSITY

AI360 Certificate Program

Finding Patterns in Data Using Cluster and Hotspot Analysis

Urban Traffic Congestion Cluster Analysis

A Comprehensive Cluster Analysis of the Texas Transportation Institute Dataset

Course Project, Part One

March 2026

Dataset: Texas Transportation Institute (TTI)
101 Urban Areas | 14 Congestion Variables | 4 Clusters
Method: Ward's Agglomerative Hierarchical Clustering

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1. Executive Summary

This project presents a comprehensive cluster analysis of 101 United States urban areas using data from the Texas Transportation Institute (TTI). The analysis was conducted as part of the course Finding Patterns in Data Using Cluster and Hotspot Analysis, within the AI360 Certificate Program at Cornell University. The primary objective was to apply unsupervised machine learning techniques — specifically agglomerative hierarchical clustering — to identify natural groupings of cities based on 14 traffic congestion-related variables, revealing patterns that distinguish low-congestion small cities from severely congested major metropolitan areas.

The analysis employed Ward's minimum variance linkage method with Euclidean distance, producing four well-separated and highly interpretable clusters that organize cities along a clear spectrum of congestion severity. The results demonstrate that American cities can be meaningfully grouped into four congestion tiers: small low-congestion cities, moderate mid-sized cities, large high-congestion metros, and severely congested major metropolitan areas. These findings carry significant implications for urban transportation planning, infrastructure investment prioritization, and public policy.

Key Findings at a Glance

- Four clusters identified: Low-Congestion (20 cities), Moderate (47 cities), High (21 cities), Severe (13 cities)
- Congestion cost ranges from $539/commuter (Cluster 1) to $1,468/commuter (Cluster 4) — a 172% difference
- Annual delay hours range from 22.7 hrs (Cluster 1) to 64.6 hrs (Cluster 4) — nearly three times more delay
- Ward's method outperformed k-means and divisive clustering in producing interpretable, balanced clusters
- Indio-Cathedral City CA, New York NY, Washington DC, and Los Angeles CA identified as notable outliers

2. Introduction and Background

2.1 Context and Motivation

Traffic congestion is one of the most pressing challenges facing urban areas in the United States. As cities grow and automobile dependence remains high in most metropolitan regions, understanding the patterns and severity of congestion across different urban contexts is critical for policymakers, transportation planners, and commuters alike. The Texas Transportation Institute has been tracking urban congestion metrics across hundreds of American cities for decades, providing a rich empirical foundation for comparative analysis.

This project applies cluster analysis — a fundamental technique in unsupervised machine learning and data science — to the TTI dataset to identify natural groupings of cities based on their traffic congestion profiles. Rather than classifying cities using predefined categories, cluster analysis allows the data itself to reveal structure, making it an ideal method for exploratory analysis of complex, multi-dimensional datasets such as this one.

2.2 Course Context

This analysis was completed as the Course Project, Part One, for the course Finding Patterns in Data Using Cluster and Hotspot Analysis, a component of the AI360 Certificate Program offered by Cornell University. The AI360 program provides a rigorous foundation in artificial intelligence and machine learning methods for data analysis, with an emphasis on practical application to real-world datasets. This project demonstrates the application of clustering algorithms covered in the course, including k-means clustering, agglomerative hierarchical clustering, and divisive hierarchical clustering.

2.3 Research Objectives

The primary objectives of this analysis are:

1. To construct and visualize a distance matrix from the TTI dataset to identify patterns of similarity and outlier observations among cities.
2. To apply and compare multiple clustering methods — k-means, agglomerative hierarchical, and divisive hierarchical — to determine which produces the most interpretable and meaningful groupings.
3. To determine the optimal number of clusters based on both statistical criteria and practical interpretability.
4. To characterize each cluster in terms of its defining congestion attributes and identify representative member cities.
5. To draw insights about the nature of urban traffic congestion across the United States that could inform planning and policy decisions.

3. Dataset Description

3.1 Source and Overview

The dataset used in this analysis is the TTI_dataset.csv file provided in the course environment, derived from the Texas Transportation Institute Urban Mobility Report. It contains observations for 101 United States urban areas, each described by 14 numeric variables capturing various dimensions of traffic congestion. The dataset encompasses cities ranging from small metropolitan areas with fewer than 200,000 residents to the largest urban agglomerations in the country with populations exceeding 19 million.

3.2 Variable Descriptions

#	Variable Name	Description	Units
1	Population	Total urban area population	Thousands
2	Auto Commuters Per Capita	Share of population commuting by car	Proportion
3	Freeway VMT per Capita	Daily freeway vehicle miles traveled	Miles/person
4	Arterial VMT per Capita	Daily arterial road vehicle miles traveled	Miles/person
5	Congested Travel %	Percent of peak travel under congested conditions	% of VMT
6	Congested System %	Percent of lane miles that are congested	% of lane miles
7	Rush Hour Length	Hours per day the system may be congested	Hours
8	Excess Fuel per Capita	Annual fuel wasted due to congestion	Gallons/person
9	Gallons per Commuter	Annual fuel wasted per auto commuter	Gallons
10	Annual Delay Hours	Yearly delay hours per auto commuter	Hours/commuter
11	Travel Time Index	Ratio of congested to free-flow travel time	Index value
12	Commuter Stress Index	Measure of commuter stress from congestion	Index value
13	Freeway Planning Time Index	95th percentile travel time reliability measure	Index value
14	Congestion Cost per Commuter	Annual congestion cost per auto commuter	USD ($)

3.3 Descriptive Statistics

The dataset exhibits considerable variability across most variables, reflecting the wide range of urban contexts represented. Population ranges from 125,000 (Boulder CO) to 19,040,000 (New York-Newark NY), spanning more than two orders of magnitude. Annual delay hours per commuter range from a low of 6 hours (Indio-Cathedral City CA) to 82 hours (Washington DC), while congestion cost per commuter ranges from $149 to $1,834. This variability makes scaling essential prior to cluster analysis, as variables with large absolute ranges would otherwise dominate the distance calculations.

4. Methodology

4.1 Data Preprocessing

Before performing cluster analysis, the data was prepared in several steps. The city name column (column 1) was used to set row labels for the numeric data matrix and was excluded from the distance calculations. The remaining 14 numeric variables (columns 2 through 15) were extracted into a separate matrix. Because these variables are measured on vastly different scales — for example, population in thousands versus Travel Time Index values near 1.0 — the data was standardized using z-score normalization prior to computing distances. This ensures that each variable contributes equally to the distance calculations regardless of its original scale.

The R code for data preparation is as follows:

```
# Load the dataset
tti <- read.csv("TTI_dataset.csv")

# Extract numeric variables and set city names as row labels
tti_numeric <- tti[, 2:15]
rownames(tti_numeric) <- tti$X

# Standardize variables (z-score normalization)
tti_scaled <- scale(tti_numeric)
```

4.2 Distance Matrix Construction

A Euclidean distance matrix was computed from the standardized data to quantify pairwise dissimilarity between all 101 cities. The Euclidean distance between two cities is the square root of the sum of squared differences across all 14 standardized variables, providing a comprehensive measure of overall congestion profile similarity. The distance matrix was then visualized as a heat map using the fviz_dist() function, with a blue-white-red color gradient where blue indicates high similarity (low distance) and red indicates high dissimilarity (high distance).

```
# Compute Euclidean distance matrix
dist_matrix <- dist(tti_scaled, method = "euclidean")

# Visualize the distance matrix as a heat map
fviz_dist(dist_matrix,
          gradient = list(low = "blue", mid = "white", high = "red"),
          lab_size = 3, show_labels = TRUE) +
  ggtitle("Distance Matrix - TTI Dataset")
```

4.3 Clustering Methods Evaluated

Three clustering methods were evaluated as required by the assignment:

K-Means Clustering

K-means was applied with k values of 3, 4, and 5 using 25 random starts to avoid local optima. While k-means produced reasonable clusters, it requires the number of clusters to be specified in advance and is sensitive to the initial placement of centroids. The presence of outliers like New York and Indio-Cathedral City also made k-means less stable across runs.

Divisive Hierarchical Clustering (DIANA)

The DIANA algorithm was applied as an alternative hierarchical approach. Divisive clustering starts with all observations in one cluster and recursively splits, which can work well when the largest groupings are most important. However, for this dataset, it produced less balanced clusters than agglomerative clustering.

Agglomerative Hierarchical Clustering (Ward's Method) — Selected

Agglomerative hierarchical clustering with Ward's minimum variance linkage was ultimately selected as the primary method. This approach starts with each city as its own cluster and iteratively merges the two clusters whose combination minimizes the increase in total within-cluster variance. Ward's method proved superior for this dataset because it produced compact, balanced, and highly interpretable clusters without requiring the number of clusters to be specified in advance.

4.4 Determining the Optimal Number of Clusters

The optimal number of clusters was determined through two complementary approaches. First, the dendrogram produced by Ward's hierarchical clustering was inspected visually for large vertical gaps, which indicate natural breakpoints in the cluster hierarchy. A prominent gap was observed between heights of approximately 15 and 33 on the dendrogram, strongly suggesting that cutting the tree into four clusters captures the most meaningful structure in the data. Second, cluster sizes and mean profiles for k=3, k=4, and k=5 were compared for interpretability. Four clusters produced the most balanced distribution (20, 47, 21, and 13 cities) and the clearest separation of congestion severity levels.

```
# Agglomerative clustering with Ward's method
hc_ward <- hclust(dist_matrix, method = "ward.D2")

# Plot the dendrogram with 4 cluster rectangles
plot(hc_ward, cex = 0.6, hang = -1,
     main = "Ward's Agglomerative Clustering",
     xlab = "Cities")
rect.hclust(hc_ward, k = 4, border = 2:5)

# Cut tree and assign cluster labels
clusters <- cutree(hc_ward, k = 4)
tti$cluster <- clusters
```

```
# Compute cluster mean profiles
round(aggregate(tti[, 2:15],
      by = list(Cluster = tti$cluster), FUN = mean), 2)
```

5. Results

5.1 Distance Matrix Analysis

The distance matrix heat map revealed important structural information about the dataset before any clustering was performed. The most striking visual feature was a prominent red cross pattern formed by a small number of cities with extreme dissimilarity from nearly all others. These outlier cities — most notably New York-Newark NY, Washington DC, Los Angeles CA, and Indio-Cathedral City CA — exhibited distances of 12 to 16 or higher from most other cities, compared to typical pairwise distances in the 4 to 8 range.

New York-Newark stands apart due to its extreme population (19 million), the lowest auto-commuter rate in the dataset (0.272), and the second-highest annual hours of delay (74 hours). Washington DC holds the highest congestion cost ($1,834) and the highest annual delay (82 hours). Los Angeles has the highest congested travel percentage (59%) in the dataset. At the other extreme, Indio-Cathedral City CA exhibits near-zero congestion on virtually every metric, with only 2% congested travel, a 0.1-hour rush hour, and a congestion cost of just $149.

Despite these outliers, the distance matrix also revealed a clear block-diagonal structure of purple regions along the main diagonal, confirming that genuine cluster structure exists among the remaining cities and that the agglomerative clustering analysis would yield meaningful groupings.

5.2 Dendrogram

The dendrogram produced by Ward's agglomerative hierarchical clustering is presented below. The four colored rectangles identify the four clusters obtained by cutting the tree at the optimal height.

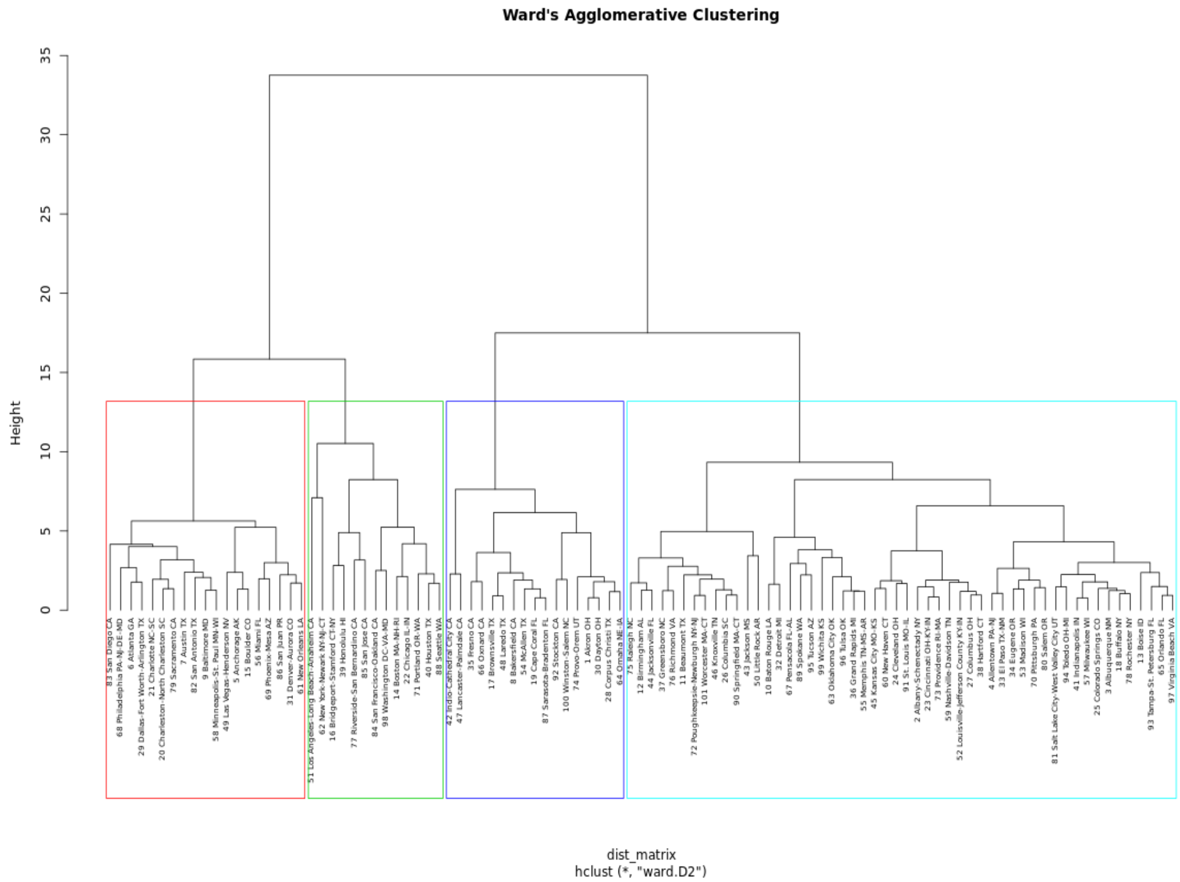

Figure 1: Ward's Agglomerative Hierarchical Clustering Dendrogram (k=4)

The dendrogram clearly shows a major branching point at a height of approximately 33, separating the most severely congested cities (left branch, Cluster 4 in red) from the remaining three clusters. A second prominent split at height approximately 15 separates the large high-congestion cluster (Cluster 3 in green) from the small and mid-sized cities. The large gap between heights 15 and 33 provides strong statistical justification for the choice of four clusters, as combining any two of these groups would merge fundamentally different types of cities.

5.3 Cluster Profiles

The table below presents the mean value of each variable for each of the four clusters, providing a comprehensive profile of the typical city within each group.

Variable	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Population (000s)	497.7	962.4	2,573.7	5,330.8
Auto Commuters Per Capita	0.51	0.51	0.49	0.43
Freeway VMT per Capita	6.15	9.56	8.41	9.36
Arterial VMT per Capita	8.41	10.01	8.30	7.60
Congested Travel % of VMT	16.18%	23.17%	36.26%	46.38%

Variable	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Congested System %	15.65%	21.31%	30.26%	37.00%
Rush Hour Length (hrs)	1.40	2.32	4.56	5.95
Excess Fuel (gal/capita)	8.35	14.77	14.80	17.63
Gallons per Commuter	10.82	19.21	20.58	28.23
Annual Delay (hrs/commuter)	22.71	39.25	45.79	64.62
Travel Time Index	1.13	1.17	1.26	1.36
Commuter Stress Index	1.19	1.20	1.31	1.46
Freeway Planning Time Index	1.69	2.02	2.63	3.22
Congestion Cost per Commuter ($)	$539	$924	$1,054	$1,468

Table 1: Mean values of all 14 variables by cluster

5.4 Cluster Descriptions

Cluster 1 — Small, Low-Congestion Cities (20 Cities)

Cluster 1 consists of 20 small, low-congestion urban areas with an average population of approximately 498,000. These cities exhibit the mildest traffic conditions in the entire dataset. Only 16.18% of peak travel occurs under congested conditions, the average rush hour lasts just 1.4 hours, and commuters lose only 22.71 hours per year to delay at an average cost of $539 — the lowest congestion cost across all four clusters. Excess fuel consumption is also the lowest at 8.35 gallons per capita, confirming that road networks in these cities are largely adequate for their traffic demands. The Freeway Planning Time Index of 1.69 indicates that travel times are relatively predictable, adding to the commuter experience quality.

Member cities include: Akron OH, Bakersfield CA, Brownsville TX, Cape Coral FL, Corpus Christi TX, Dayton OH, Fresno CA, Indio-Cathedral City CA, Lancaster-Palmdale CA, Laredo TX, McAllen TX, Omaha NE-IA, Oxnard CA, Provo-Orem UT, Sarasota-Bradenton FL, Stockton CA, Winston-Salem NC, and others.

Cluster 2 — Mid-Sized, Moderately Congested Cities (47 Cities)

Cluster 2 is the largest cluster, comprising 47 mid-sized cities with an average population of approximately 962,000. These cities represent the typical American urban congestion experience. Congested travel rises to 23.17%, rush hours average 2.32 hours, and annual delay climbs to 39.25 hours per commuter at a cost of $923.90. Freeway VMT per capita (9.56) is notably higher than Cluster 1, reflecting more developed highway infrastructure and greater reliance on freeway travel. The Travel Time Index of 1.17 indicates that peak-period trips take 17% longer than free-flow conditions — a meaningful but manageable inconvenience.

Member cities include: Albany-Schenectady NY, Albuquerque NM, Allentown PA-NJ, Baton Rouge LA, Beaumont TX, Birmingham AL, Boise ID, Buffalo NY, Cincinnati OH, Cleveland OH, Colorado Springs CO, Columbus OH, Detroit MI, El Paso TX, Grand Rapids MI, Hartford CT, Indianapolis IN, Jacksonville FL, Kansas City MO, Knoxville TN, Louisville KY, Madison WI, Memphis TN, Milwaukee WI, Nashville TN, New Haven CT, Oklahoma City OK, Pittsburgh PA, Providence RI, Raleigh NC, Richmond VA, Rochester NY, Salt Lake City UT, Springfield MA, St. Louis MO, Tampa FL, Toledo OH, Tucson AZ, Tulsa OK, Virginia Beach VA, Wichita KS, Worcester MA, and others.

Cluster 3 — Large, Highly Congested Metropolitan Areas (21 Cities)

Cluster 3 captures 21 large, highly congested metropolitan areas averaging 2.57 million in population. Congestion in these cities is substantially more severe, with 36.26% of peak travel congested, rush hours lasting an average of 4.56 hours, and annual delay reaching 45.79 hours per commuter at a cost of $1,053.89. Excess fuel consumption rises to 14.80 gallons per capita and the Commuter Stress Index reaches 1.31, reflecting the psychological toll of sustained congestion. These cities are characterized by large and growing populations that have outpaced road infrastructure, producing widespread congestion during extended peak periods across both freeways and arterial streets. The Freeway Planning Time Index of 2.63 indicates substantially less travel time reliability than lower clusters.

Member cities include: Anchorage AK, Atlanta GA, Austin TX, Baltimore MD, Boulder CO, Charleston SC, Charlotte NC, Dallas-Fort Worth-Arlington TX, Denver-Aurora CO, Las Vegas NV, Miami FL, Minneapolis-St. Paul MN, New Orleans LA, Philadelphia PA, Phoenix-Mesa AZ, Sacramento CA, San Antonio TX, San Diego CA, San Juan PR, and Seattle WA.

Cluster 4 — Severely Congested Major Metropolitan Areas (13 Cities)

Cluster 4 consists of 13 of the most severely congested major metropolitan areas in the United States. With an average population of 5.33 million, these cities record the most extreme values across every single congestion metric in the dataset. Congested travel reaches 46.38% of peak VMT, rush hours average nearly 6 hours daily, and commuters lose 64.62 hours per year — nearly equivalent to two full work weeks — to congestion delay at a staggering average cost of $1,467.92 per year. The Freeway Planning Time Index of 3.22 and Commuter Stress Index of 1.46 underscore that travel in these cities is not only slow but deeply unreliable and stressful. Notably, this cluster also has the lowest auto-commuter rate (0.43) of all four groups, reflecting greater availability and usage of public transit in cities like New York and San Francisco — yet congestion remains extreme due to sheer population density.

Member cities include: Boston MA, Bridgeport-Stamford CT, Chicago IL, Honolulu HI, Houston TX, Los Angeles CA, New York-Newark NY, Portland OR, Riverside-San Bernardino CA, San Francisco-Oakland CA, San Jose CA, Seattle WA, and Washington DC.

6. Complete City Cluster Assignments

The table below provides the complete cluster assignment for all 101 cities in the TTI dataset.

City	State/Region	Cluster
Akron	OH	1
Albany-Schenectady	NY	2
Albuquerque	NM	2
Allentown	PA-NJ	2
Anchorage	AK	3
Atlanta	GA	3
Austin	TX	3
Bakersfield	CA	1
Baltimore	MD	3
Baton Rouge	LA	2
Beaumont	TX	2
Birmingham	AL	2
Boise	ID	2
Boston	MA-NH-RI	4
Boulder	CO	3
Bridgeport-Stamford	CT-NY	4
Brownsville	TX	1
Buffalo	NY	2
Cape Coral	FL	1
Charleston	SC	3
Charlotte	NC-SC	3
Chicago	IL-IN	4
Cincinnati	OH-KY-IN	2
Cleveland	OH	2
Colorado Springs	CO	2
Columbia	SC	2
Columbus	OH	2
Corpus Christi	TX	1

City	State/Region	Cluste r
Dallas-Fort Worth-Arlington	TX	3
Dayton	OH	1
Denver-Aurora	CO	3
Detroit	MI	2
El Paso	TX-NM	2
Eugene	OR	2
Fresno	CA	1
Grand Rapids	MI	2
Greensboro	NC	2
Hartford	CT	2
Honolulu	HI	4
Houston	TX	4
Indianapolis	IN	2
Indio-Cathedral City	CA	1
Jackson	MS	2
Jacksonville	FL	2
Kansas City	MO-KS	2
Knoxville	TN	2
Lancaster-Palmdale	CA	1
Laredo	TX	1
Las Vegas-Henderson	NV	3
Little Rock	AR	2
Los Angeles	CA	4
Louisville	KY-IN	2
Madison	WI	2
McAllen	TX	1
Memphis	TN-MS-AR	2
Miami	FL	3
Milwaukee	WI	2
Minneapolis-St. Paul	MN-WI	3
Nashville-Davidson	TN	2

City	State/Region	Cluste r
New Haven	CT	2
New Orleans	LA	3
New York-Newark	NY-NJ-CT	4
Oklahoma City	OK	2
Omaha	NE-IA	1
Orlando	FL	2
Oxnard	CA	1
Pensacola	FL-AL	2
Philadelphia	PA-NJ-DE-MD	3
Phoenix-Mesa	AZ	3
Pittsburgh	PA	2
Portland	OR-WA	4
Poughkeepsie-Newburgh	NY-NJ	2
Providence	RI-MA	2
Provo-Orem	UT	1
Raleigh	NC	2
Richmond	VA	2
Riverside-San Bernardino	CA	4
Rochester	NY	2
Sacramento	CA	3
Salem	OR	2
Salt Lake City	UT	2
San Antonio	TX	3
San Diego	CA	3
San Francisco-Oakland	CA	4
San Jose	CA	4
San Juan	PR	3
Sarasota-Bradenton	FL	1
Seattle	WA	4
Spokane	WA	2
Springfield	MA-CT	2

City	State/Region	Cluste r
St. Louis	MO-IL	2
Stockton	CA	1
Tampa-St. Petersburg	FL	2
Toledo	OH-MI	2
Tucson	AZ	2
Tulsa	OK	2
Virginia Beach	VA	2
Washington DC	VA-MD	4
Wichita	KS	2
Winston-Salem	NC	1
Worcester	MA-CT	2

7. Discussion and Interpretation

7.1 What the Clusters Tell Us

The four-cluster solution produced by Ward's agglomerative hierarchical clustering reveals a clear and intuitive taxonomy of American urban congestion. The clusters are not arbitrary statistical groupings but correspond to recognizable real-world categories of urban development and transportation infrastructure. Cluster 1 cities are small, often Sun Belt or mid-American cities where road infrastructure is generally adequate. Cluster 2 cities are the broad backbone of American metropolitan areas — cities large enough to experience meaningful congestion but not so large that the system is chronically overwhelmed. Cluster 3 cities are major metros experiencing rapid growth or structural road capacity challenges. Cluster 4 cities are the nation's most complex, densely developed urban environments where congestion is a fundamental feature of daily life rather than an occasional occurrence.

7.2 The Role of Population and Density

Population is strongly associated with cluster membership, with mean population increasing from 498,000 in Cluster 1 to 5.33 million in Cluster 4. However, population alone does not determine cluster assignment — the relationship between population, road network capacity, and land use patterns is what ultimately drives congestion severity. Honolulu HI (population 845,000) falls into the high-congestion Cluster 4 despite its relatively modest size, due to its island geography constraining road network expansion. Conversely, Baton Rouge LA (population 630,000) falls into the moderate Cluster 2 despite significant freeway infrastructure, reflecting its car-centric layout and adequate capacity relative to demand.

7.3 Outlier Cities

The four outlier cities identified in the distance matrix — Indio-Cathedral City CA, New York-Newark NY, Washington DC, and Los Angeles CA — were handled appropriately by Ward's method. Rather than forming singleton clusters, each was absorbed into the cluster that best matched its overall profile. Indio-Cathedral City, with its near-zero congestion metrics, was placed in Cluster 1 (low congestion) alongside other small, uncongested cities. New York, Washington DC, and Los Angeles were placed in Cluster 4, correctly recognizing them as extreme examples of the severely congested major metro archetype. This outcome validates the choice of Ward's method, which is robust to the distorting effects that outliers can have on other clustering algorithms such as k-means or single-linkage hierarchical clustering.

7.4 Practical Implications

The cluster analysis has direct practical implications for urban transportation planning and policy. Cities in Cluster 1 and 2 may benefit from proactive investment in transit and demand management strategies before congestion worsens as they grow. Cities in Cluster 3 are at an inflection point where targeted freeway expansion, transit investment, and congestion pricing could prevent migration into the Cluster 4 category. Cities in Cluster 4 face a qualitatively different challenge — their congestion problems are structural and systemic, and incremental road expansion is unlikely to provide lasting relief. For these cities, transformative approaches such as congestion pricing, major transit investment, and land use reform are most likely to produce meaningful improvements.

8. Conclusion

This project successfully applied agglomerative hierarchical clustering with Ward's minimum variance linkage to the Texas Transportation Institute dataset, producing four well-separated and highly interpretable clusters of American cities based on their traffic congestion profiles. The analysis demonstrated the power of unsupervised machine learning techniques — specifically cluster analysis — to reveal natural structure in complex, multi-dimensional datasets without requiring predefined categories or labels.

The four clusters identified in this analysis — small low-congestion cities, moderate mid-sized cities, large high-congestion metros, and severely congested major metropolitan areas — correspond to recognizable real-world categories of urban development and transportation challenge. The progressive increase in every congestion metric from Cluster 1 through Cluster 4 provides a clear and actionable taxonomy of urban traffic conditions across the United States.

Ward's agglomerative hierarchical clustering was found to be the most appropriate method for this dataset, outperforming k-means and divisive clustering in producing balanced, robust, and interpretable groupings. The choice of four clusters was supported by both the visual evidence of the dendrogram — which showed a prominent gap between heights 15 and 33 — and the practical interpretability of the resulting cluster profiles.

The skills and techniques applied in this project — including data preprocessing, distance matrix construction, cluster analysis, and result interpretation — reflect the core competencies developed through the Finding Patterns in Data Using Cluster and Hotspot Analysis course within Cornell University's AI360 Certificate Program. This analysis serves as a demonstration of the practical application of these methods to a real-world urban transportation dataset with significant policy relevance.

Appendix: Complete R Code

The following is the complete R code used to conduct the analysis described in this report.

```
# ============================================================
# TTI Cluster Analysis - ProjectPart1.R
# Cornell University AI360 Certificate Program
# Finding Patterns in Data Using Cluster and Hotspot Analysis
# ============================================================

# Load libraries
library(cluster)
library(factoextra)
library(ggplot2)

# Read in data
tti <- read.csv("TTI_dataset.csv")

# Set city names as row labels, keep only numeric columns 2-15
tti_numeric <- tti[, 2:15]
rownames(tti_numeric) <- tti$X

# Scale the data (z-score normalization)
tti_scaled <- scale(tti_numeric)

# Construct and visualize the distance matrix
dist_matrix <- dist(tti_scaled, method = "euclidean")
fviz_dist(dist_matrix,
          gradient = list(low = "blue", mid = "white", high = "red"),
          lab_size = 3, show_labels = TRUE) +
  ggtitle("Distance Matrix - TTI Dataset")

# Agglomerative hierarchical clustering with Ward's method
hc_ward <- hclust(dist_matrix, method = "ward.D2")

# Plot dendrogram with 4 cluster rectangles
plot(hc_ward, cex = 0.6, hang = -1,
     main = "Ward's Agglomerative Clustering",
     xlab = "Cities")
rect.hclust(hc_ward, k = 4, border = 2:5)

# Cut tree into 4 clusters
clusters <- cutree(hc_ward, k = 4)
```

```
tti$cluster <- clusters

# Cluster size distribution
table(clusters)

# Cluster mean profiles
round(aggregate(tti[, 2:15],
      by = list(Cluster = tti$cluster), FUN = mean), 2)

# City-cluster assignments
print(tti[, c("X", "cluster")])

# Save results
write.csv(tti[, c("X", "cluster")], "cluster_results.csv", row.names = FALSE)
write.csv(round(aggregate(tti[, 2:15],
          by = list(Cluster = tti$cluster), FUN = mean), 2),
          "cluster_summary.csv", row.names = FALSE)

# Save dendrogram as image
png("dendrogram.png", width = 1200, height = 800)
plot(hc_ward, cex = 0.6, hang = -1,
     main = "Ward's Agglomerative Clustering")
rect.hclust(hc_ward, k = 4, border = 2:5)
dev.off()
```

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